

# TailgateAlert System Overview

*A simple overview of how TailgateAlert detects, sends, warns, and saves alerts.*

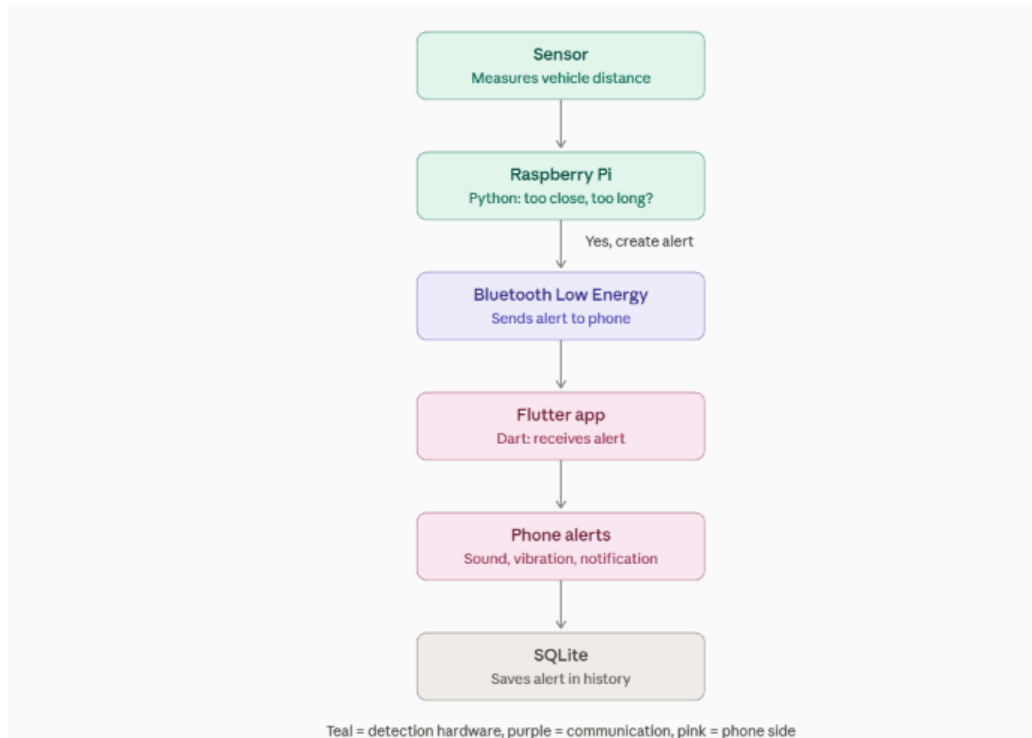


Figure 1. TailgateAlert system flow

## How It Works

1. **Sensor:** Measures how far away the vehicle behind you is.

1. The sensor repeatedly measures the distance to the vehicle behind us.
2. The Raspberry Pi compares that measurement with a configurable distance threshold.
3. If the vehicle stays inside that threshold, the Pi starts measuring how long it remains there.
4. The duration is calculated as the current time minus the time the close-following condition started.
5. If both the distance and duration conditions are met, the system creates an alert and sends it to the phone.

2. **Raspberry Pi:** Runs Python code that checks if the vehicle is too close for too long. If yes, it creates an alert.

3. **Bluetooth Low Energy (BLE):** Sends the alert from the Raspberry Pi to the phone.

4. **Flutter App:** Built with Flutter and Dart. It receives the Bluetooth alert.

5. **Phone Alert:** Plays a sound, vibrates, and shows a notification.

6. **SQLite:** Saves the alert on the phone so it can be viewed later in History.

## Code Idea

Raspberry Pi (Python)

```
distance = 6.4
time_close = 3.2
```

```
if distance < 8 and time_close >= 3:  
    print("Send tailgating alert")
```

**Meaning:** If the car is closer than 8 meters for at least 3 seconds, the Pi sends an alert.

### Phone App (Flutter/Dart)

```
void receiveAlert() {  
    playSound();  
    vibrate();  
    showNotification();  
    saveToHistory();  
}
```

**Meaning:** When an alert arrives, the phone warns the driver and saves the event.

## How to Run It

**Raspberry Pi:** `python3 tailgate_alert.py`

**Flutter development:** `flutter run`

**After installation:** Open the TailgateAlert app. It scans for the Pi, connects through BLE, and listens for alerts.